Lab Exercises 6 – CSS Responsive Layout and Accessibility

Aims

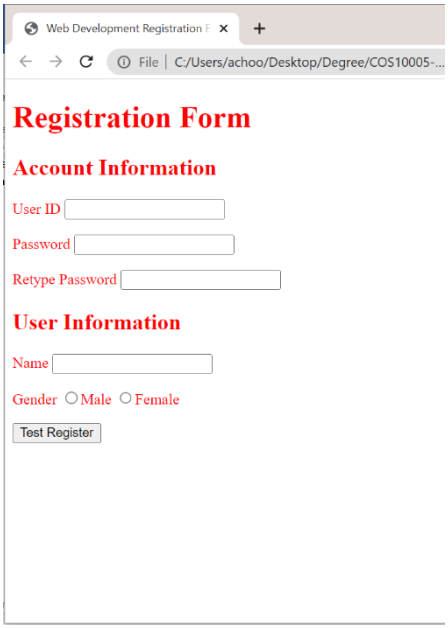
- To learn how to specify and apply different CSS rules for different devices, e.g., desktops and mobile phones;
- To get familiar with Web Accessibility Initiative;
- To learn how to test web pages for compliance and improve the quality of web pages, and introduce some assistive technologies.

Task 1: CSS Responsive Layout

We are going to create two (2) CSS files for one (1) registration form. One will be used for a desktop device and the other for a mobile device.

Step 1:

- Use notepad++ to open and edit regform.html.
- Format the regform.html to look like in desktop and a mobile phone like below:



a) Desktop View



b) Mobile View

Step 2:

- Use notepad++ to create two (2) CSS files: desktop.css and mobile.css, that will present the forms shown above.
- To apply the two (2) CSS files to regform.html, you need to add the following code to the <head> part of regform.html.

```
<link href="desktop.css" rel="stylesheet" media="screen and
(minwidth:768px)" />
<link href="mobile.css" rel="stylesheet" media="screen and
(maxwidth:480px)" />
```

- By doing so, desktop.css will be applied to the regform.html when the width of the browser display regform.html is greater than 768 pixels, which is usually the minimum width of a PC monitor. If the width of the browser is lower than 480 pixels, mobile.css will be applied to regform.html, displayed the web page in mobile-friendly manner, e.g., larger font and compact layout.

Step 3:

- Specify only one (1) rule in desktop.css:

```
body {
color:red;
}
```

- Using this CSS rule, when all the text on the web page are rendered to red. Now you can see the desktop.css is taking effect.

Step 4:

- Now, let's working on mobile.css. Follow the instruction below to complete and apply CSS rules to present regform.html in mobile view. You might need to specify the id or class attributes of applicable HTML elements so that they can be properly selected in the CSS file for CSS application.
    - Increase the font size in the **form** to 150%;
    - Set the font color to blue;
    - Format heading 1 style as below:
        - Center the text
    - Format heading 2 style as below:
        - Change the font size to 40 pixels
        - Remove the bottom margin
    - Format <input> elements as below:
        - Make all elements occupy the entire width
    - Format <p> elements as below:
        - Remove all margins around
    - Format radio buttons as below:
        - Set width to 15%

- Format submit buttons as below:
  - Set width to 40%
  - Center the button (you may look into)

Step 5:

- Instead of two (2) css files, merge desktop.css and mobile.css into one (1) css file: style.css.
- Add the following CSS rule to style.css:

```
@media screen and (min-width: px) {

}
```

Copy the script from desktop.css and paste it here.

Indicate the right px for the min-width and max-width

```
@media screen and (max-width: px) {

}
```

Copy the script from mobile.css and paste it here.

- Remark the following codes in regform.html.

```
<link href="desktop.css" rel="stylesheet" media="screen and
(minwidth:768px)" />
<link href="mobile.css" rel="stylesheet" media="screen and
(maxwidth:480px)" />
```

- Now add the following code to the <head> part of regform.html to replace the previous 2 links to mobile.css and desktop.css.

```
<link href="style.css" rel="stylesheet" />
```

- Now only one (1) CSS file is applied to regform.html. based on the width of the browser window, different CSS rules will kick in and present regform.html into two (2) different ways. Test it and see for yourselves.

Step 6: Validate the page and fix any errors displayed and revalidate.

- Use the validator at http://validator.w3.org to validate your html file.
- Use the CSS validator at http://jigsaw.w3.org/css-validator/ .